higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CHEMISTRY
NQF LEVEL 3

14 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 1.1.1 6% (1)

1.1.2
- Nitrogen improves root growth.
- It also helps with the uptake and absorption of other essential nutrients. (2)

1.1.3
- Nitrate (NO_3^-)
- Ammonium (NH_4^+) (2)

1.1.4 It needs to be in contact with other chemicals such as oxygen and hydrogen before it is absorbed by the plant's root system. (2)

1.1.5 Plants that lack potassium develop leaf scorch,✓ the edges of the leaves become yellow✓ and if the deficiency continues, the leaves become brown and eventually dry out and appear to be dead.✓ (3)

1.2 1.2.1 K_2SO_4
1.2.2 $NH_4H_2PO_4$
1.2.3 $Ca(H_2PO_4)_2$
(3 × 2) (6)

1.3
- Sulphur extraction
- Gas cleaning
- Sulphur trioxide formation
- Catalysed formation of oleum
- Dissociation of oleum to form sulphuric acid (5)

1.4 $3NO_2(g) + H_2O(\ell) \rightarrow 2HNO_3(aq) + NO(g)$ (4)
[25]

QUESTION 2

2.1 2.1.1 Na_2CO_3
2.1.2 Na_2SO_4
2.1.3 NaOH
2.1.4 $NaHCO_3$
2.1.5 $NaC\ell$
(5 × 1) (5)

2.2 2.2.1 Calcareous✓ and argillaceous rocks✓ (2)

2.2.2 The dry process is more difficult to control. (1)

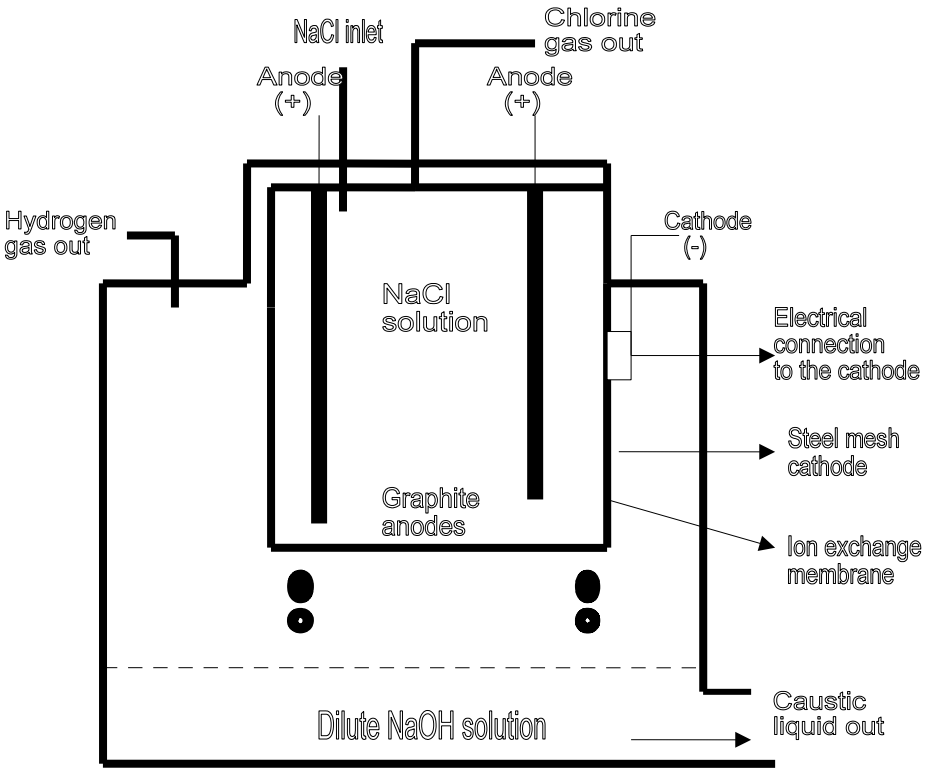
2.2.3 Slagment consists of a mixture of slag✓ and hydrated lime✓ and is a by-product of the iron smelting process.✓ (3)

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2.2.4
- It gives a strong final product
- It provides better resistance to chemicals like sulphates and chlorides
- It is easier to work with
- It has a better colour
- It provides a better finish
- It provide better resistance to water damage (Any 5 × 1) (5)

2.3 A – Limestone
B – Kiln
C – Slaker
D – Settling tank
E – Slaked lime
F – Water
G – Mill
H – Silo
I – Carbon dioxide (9)
[25]

QUESTION 3



[10]

QUESTION 4

4.1 4.1.1 Hard water is water that contains dissolved salts of magnesium and calcium compounds and does not form lather with soap. (2)

4.1.2 Surfactants change the properties of the surface of a liquid by lowering its surface tension.✓ They are formed by long hydrocarbons with one hydrophobic (repels water and is attracted to dirt) one hydrophilic (attracted to water).✓ When a detergent is used to wash an object, the hydrophobic end attaches to dirt✓ while the hydrophilic end pulls the chain into the water and along with it the dirt particle.✓ In this way dirt is lifted off the object and it is suspended in water.✓ (5)

4.1.3
- Bubble bath
- Shampoos (2)

4.2 Fillers are cheap substances that are added to detergents to increase the quantity or the bulk of the detergent. Fillers also prevent corrosion and adjusts the pH of the detergents. (2)

4.3
- Sodium or potassium hydroxide (caustic soda)
- Animal or vegetable fats and oils
- Water (3)

4.4
- Saponification
- Lye separation (spent lye)
- Soap washing
- Lye separation (half spent lye)
- Neutralisation
- Drying (6)

[20]

QUESTION 5

5.1
- Hardwood (angiosperms)
- Softwood (gymnosperms) (2)

5.2 5.2.1 It is the outer surface of the tree trunk, does not contain any useful fibres.

5.2.2 The formation of new layers of wood produced during each growing season takes place here.

5.2.3 It is found underneath the cambium in the form of rings and it is the main source of fibre for pulp.

5.2.4 It is found underneath the sapwood and is made up of dead wood cells that do not contain any useful fibres.

(4 × 2) (8)

5.3 A – Bleaching chemicals
B – Bleaching
C – Pulp
D – Bleached pulp
E – Kraft pulp
F – Washing and screening
G – Chips
H – White liquor
I – Recausticising
J – Drying on paper machines (10)
[20]

TOTAL: 100